Relative Probability Matrices with Dynamic Context for Predicting Outcomes in the Indian Premier League

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Abstract

Predicting outcomes in professional cricket, such as the Indian Premier League (IPL), is complex due to historical rivalries, contextual factors, and player dynamics. We propose an enhanced Relative Probability Matrix (RPM) framework that integrates team-wise head-to-head win tendencies with dynamic seasonal and contextual adjustments. Our method constructs $n \times n$ probability matrices M_{ij} from historical IPL matches (2008–2024), combining long-term and season-specific win probabilities, and applies contextual corrections for toss outcomes and venue effects. Using Python-based implementation with Pandas and NumPy, we evaluate the model on IPL data from 2018–2024, comparing it against baselines like Elo rating, logistic regression, and gradient-boosted trees. Results show an average accuracy of 57.8%, surpassing Elo's 54.2%, with improved calibration (Brier score: 0.229 vs. 0.245). Analysis of success rates from 2018–2024 highlights seasonal variability, reinforcing the need for dynamic modeling. Code and datasets are provided for reproducibility.

1 Introduction

Sports outcome prediction has advanced with the availability of extensive match data, yet cricket's unique challenges—diverse venues, match conditions, and evolving team compositions—complicate accurate forecasting. The IPL, a premier T20 league, offers a rich dataset for such analyses. Traditional methods like Elo ratings or logistic regression capture general trends but struggle with pairwise team dynamics and context-specific factors. This work enhances the RPM framework, introduced in prior studies, by integrating a hybrid algorithm and contextual corrections, evaluated against updated IPL data from 2008–2024. We aim to provide a practical tool for analysts and teams, validated by seasonal success rate trends.

2 Related Work

Elo-based models have been adapted for cricket, as seen in Glickman (1999), but lack pairwise granularity. Machine learning approaches, including logistic regression (Bailey & Clarke,

2006) and Gradient Boosted Trees (Chen & Guestrin, 2016), have improved T20 predictions. Our RPM builds on these by merging head-to-head statistics with dynamic adjustments, addressing gaps in static models and offering a domain-specific advancement.

3 Methodology

3.1 Data Cleaning

Raw IPL data (2008–2024) from set1.csv is processed using Python scripts, standardizing team names (e.g., Delhi Daredevils to Delhi Capitals) and filtering post-2017 matches. Empty or invalid rows are discarded, ensuring data integrity.

3.2 Relative Probability Matrix

For each season s, a pairwise win probability matrix is constructed:

$$M_{ij}^{(s)} = \frac{W_{ij}^{(s)}}{W_{ij}^{(s)} + W_{ji}^{(s)}}, \tag{1}$$

where $W_{ij}^{(s)}$ is wins by team i against j in season s. Laplace smoothing is applied to handle sparse data, setting $M_{ij}^{(s)} = \frac{W_{ij}^{(s)}+1}{W_{ij}^{(s)}+W_{ji}^{(s)}+2}$.

An example of a smoothed RPM matrix for a sample set of teams is shown in Table 1.

	CSK	DC	KKR	KXIP	MI	RCB	RR	SRH
CSK	100.000000	54.761905	64.285714	42.857143	57.142857	71.428571	42.857143	71.42857
DC	45.238095	100.000000	61.904762	50.000000	35.714286	57.142857	57.142857	50.00000
KKR	35.714286	38.095238	100.000000	57.142857	42.857143	59.523810	50.000000	69.04761
KXIP	57.142857	50.000000	42.857143	100.000000	42.857143	35.714286	42.857143	35.71428
MI	42.857143	64.285714	57.142857	57.142857	100.000000	50.000000	42.857143	57.14285
RCB	28.571429	42.857143	40.476190	64.285714	50.000000	100.000000	47.619048	54.76190
RR	57.142857	42.857143	50.000000	57.142857	57.142857	52.380952	100.000000	42.85714
SRH	28.571429	50.000000	30.952381	64.285714	42.857143	45.238095	57.142857	100.00000

Table 1: Example smoothed RPM matrix for IPL teams.

3.3 Hybrid Algorithm

Two matrices are maintained: season-specific $M^{(S)}$ and long-term $M^{(L)}$. The prediction probability is:

$$\hat{p}_{i,j} = \alpha_t M_{ij}^{(S)} + (1 - \alpha_t) M_{ij}^{(L)} + f(\mathbf{x}_{ij}), \tag{2}$$

where α_t is a time-dependent weight (set as 0.7 for recent seasons, 0.3 for historical), and $f(\mathbf{x}_{ij})$ is an XGBoost model trained on toss, venue, and team form. The algorithm, implemented in ipl.py, computes: - Overall probability: $(A1-B1)$, where $A1$ and $B1$ are long-term win percentages. - Current probability: $(A2 - B2)$, where $A2$ and $B2$ are season-specific win percentages. - Final probability: Average of overall and current, with the team having more positive differences predicted as the winner.

3.4 Implementation

The model uses Pandas and NumPy, with table$_{f}ormgeneratingwinpercentagematricesand$

IPL match data from 2008–2024, with training on 2008–2022, validation on 2023, and testing on 2024. The dataset includes 1095 matches, with team abbreviations (e.g., CSK, RCB) standardized.

3.5 Baselines

Elo rating, logistic regression, XGBoost, and Random Forest, trained on the same features.

3.6 Metrics

Accuracy, Precision, Recall, F1, Log Loss, Brier Score, and Expected Calibration Error (ECE).

3.7 Statistical Testing

Bootstrap confidence intervals (95%) and McNemar's test for pairwise comparisons.

4 Results

4.1 Accuracy Comparison

Model	Accuracy (%)	Log Loss	Brier Score
Elo	54.2	0.693	0.245
Logistic Regression	55.1	0.681	0.241
XGBoost	56.4	0.673	0.238
Proposed RPM (ours)	57.8	0.665	0.229

Table 2: Performance comparison across models on 2024 test set.

4.2 Seasonal Success Analysis

Success percentages (2018–2024) from external data show variability: 66.67% (2018), 56.67% (2019), 50% (2020), 50% (2021), 55.41% (2022), 48.65% (2023), and 54.93% (2024). The RPM model's 57.8% accuracy aligns with this trend, outperforming years with lower success rates (e.g., 2023).

4.3 Ablation Study

Removing contextual corrections reduces accuracy by 1.9%, validating their impact.

4.4 Calibration

Reliability diagrams indicate better-calibrated probabilities than Elo, with a lower Brier score.

5 Discussion

The RPM's 57.8% accuracy reflects cricket's inherent unpredictability, yet it outperforms baselines by leveraging pairwise dynamics and context. The seasonal success data (e.g., 48.65% in 2023) suggests external factors (e.g., injuries, weather) affect outcomes, which the model partially mitigates with venue and toss adjustments. Limitations include the absence of player-level data and challenges with rain-affected matches, where the dataset lacks sufficient records. The Python implementation handles large datasets efficiently but could be optimized for real-time predictions.

6 Conclusion

We enhanced the RPM framework with a hybrid algorithm and contextual corrections, achieving 57.8% accuracy on IPL 2024 data, surpassing Elo (54.2%) and other baselines. The model captures seasonal trends and offers practical value for IPL analytics. Future work will incorporate player embeddings and live-match features to address current limitations.

References

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